



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure and Syllabus

(From Academic Session 2024-25 onwards)

B.TECH
Electrical and Electronics Engineering (EEE)

3rd SEMESTER



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY

Course Structure (From Academic Session 2024-25 onwards)

B.Tech 3rd Semester: Electrical and Electronics Engineering

Semester III

Sl. No.	Sub-Code	Subject	Hours per Week			Credit	Marks	
			L	T	P	C	CE	ESE
Theory								
1	EEE241301	Semiconductor Devices and Circuits	3	0	0	3	30	70
2	EE241302	Electrical Machines-I	3	0	0	3	30	70
3	EE241303	Digital Electronics	3	0	0	3	30	70
4	EE241304	Electrical Circuit Analysis	3	1	0	4	30	70
5	MA241305C	Mathematics III-C	3	0	0	3	30	70
6	HS241306	Indian Knowledge Systems	2	0	0	2	-	50
7	AC241307	Language other than MIL	1	1	0	0 (PP/NP)	-	50
Practical								
1	EEE241311	Basic Electronics Lab	0	0	2	1	15	35
2	EE241312	Electrical Machines-I Lab	0	0	2	1	15	35
3	EE241313	Digital Electronics Lab	0	0	2	1	15	35
4	SI241321	Social Internship	0	0	4	2	-	50
TOTAL			18	2	10	23	195	605
Total Contact Hours per week: 30								
Total Credits: 23								

N.B. AC241307 is a Mandatory Audit Course (No Credit). It will be evaluated as PP (Pass) or NP (Not Pass)

DETAILED SYLLABUS

Course Code	Course Title	Hours per week L-T-P	Credit C
EEE241301	Semiconductor Devices and Circuits	3-0-0	3

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Explain the behaviour of semiconductor devices.
- **CO2:** Analyze the functioning of various solid- state devices, including diodes, BJTs and FETs.
- **CO3:** Illustrate the biasing techniques for BJTs and FETs.
- **CO4:** Realize simple amplifier circuits using BJTs and FETs.
- **CO5:** Explain the behaviour of special semiconductor devices.

MODULE 1: Overview of the physics in Semiconductor: (6 Hours)

Review of Quantum Mechanics, Electrons in periodic Lattices, E-k diagrams; Energy bands in intrinsic and extrinsic silicon; Carrier transport: diffusion current, drift current, mobility and resistivity; sheet resistance, design of resistors; Generation and recombination of carriers; Poisson and continuity equation.

MODULE 2: P-N junction: (10 Hours)

The open circuited junction, space charge region, the biased p-n junction, I-V characteristics, Piece-wise linear characteristics & equivalent circuits, Diode resistance, Capacitance, switching time, small signal switching models; Avalanche breakdown, Zener breakdown. Half wave and Full wave single phase rectifiers, filtering and their analysis, peak inverse voltage, voltage multiplier circuits, Clipping and Clamping circuits. Zener diode and voltage regulator, Schottky diode, Varactor diode.

MODULE 3: Bipolar Junction Transistor: (12 Hours)

PNP and NPN junction transistors, various configurations of BJT and their input & output characteristics, different modes of operation, Ebers-Moll representation, Early effect, Avalanche breakdown & Punch through BJT biasing: Operating Point, DC & AC load lines, analysis of different biasing circuits and problems, Stabilization, Thermal runaway and thermal stability, BJT as a switch and amplifier.

MODULE 4: Field Effect Transistor: (8 Hours)

Structure and operation of JFET, the drain and transfer characteristics, Biasing of FETs. MOSFET: Structures and operations of Depletion and Enhancement types of MOSFET, the drain and transfer characteristics, MOS capacitor, C-V characteristics, introduction to CMOS

MODULE 5: Special Semiconductor Devices: (4 Hours)

Introduction to Photonic devices: LED, LASER diodes, Photo detectors, Solar Cells, Introduction to Power Semiconductor Devices: Power BJT, Power MOSFET, UJT, CCD, Thyristor, Tunnel diode.

Textbooks/References:

1. G. Streetman, and S. K. Banerjee, "Solid State Electronic Devices," 7th edition, Pearson, 2014.
2. D. Neamen, D. Biswas, "Semiconductor Physics and Devices," McGraw-Hill Education

3. S. M. Sze and K. N. Kwok, "Physics of Semiconductor Devices," 3rd edition, John Wiley & Sons, 2006. A.S. Sedra and K.C. Smith, Microelectronic Circuits, Saunder's College Publishing, 1991.
4. NPTEL course as suggested by the department.

Course Code	Course Title	Hours per week L-T-P	Credit C
EE241302	Electrical Machines-I	3-0-0	3

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Apply the principles of electromechanical energy conversion and electromagnetic induction to electrical machines
- **CO2:** Explain the construction and principles of operation of D.C. Generators
- **CO3:** Explain the principles of operation and control of D.C. Motors
- **CO4:** Explain the construction, characteristics and operation of transformers

MODULE 1: Electromagnetic Induction and Electromechanical energy conversion :(3 Hours)

(i) Electromagnetic Induction

Review of magnetic circuits, electromagnetic induction, and magnetically coupled circuits.

(ii) Electromechanical energy conversion

Basic principles of energy conversion field magnets, dynamically induced e.m.f. and torque in rotating machines.

MODULE 2: DC Generators and DC Motors(12 + 12 Hours)

(i) DC Generators

Constructional features, details of armature windings, methods of excitations-shunt, series and compound. Emf equation, commutation, armature reaction, inter-poles and compensatory windings, characteristics of generators, efficiency and regulation, parallel operation.

(ii) DC Motors

Torque equation of motors, speed and torque characteristic curves of shunt, series and compound motors; starting of DC motor, starters and grading of starting resistance; speed control; conventional methods and solid-state control, losses and efficiency, Swinburne's test, back-to-back test, retardation test, brake test; choice of motors for different applications.

MODULE 3: Transformers :(13 Hours)

Principles of operation of transformer, voltage and current ratios, construction – shell type and core type, single phase and poly phase; Cooling methods; e.m.f and output equation, Magnetic circuit, leakage flux and leakage reactance, phasor diagrams, per unit values of resistance and reactance, open circuit and short circuit tests, back to back test; Regulation, losses and efficiency, maximum efficiency, all-day efficiency; Auto-transformer, 3-phase transformer, Phase transformation and connections. Parallel operation of transformers, Vector grouping.

Textbooks/References:

1. Electrical Machines: Nagrath D.P. & Kothari I .J, Tata Mc Graw Hill Education, Fifth Edition, 2017.
2. Electric Machinery: Bimbra P.S, Khanna Publishers, 2021.
3. Principle of Electrical Machines: Mehta V.K. and Mehta, R., S. Chand and Co., 2019.
4. Theory of Alternating Current Machinery: Langsdorf A.S, McGraw Hill Education, 1955.
5. Electric Machinery Fundamentals: Chapman. J, McGraw Hill Book Co., Fourth Edition, 2017.
6. Electric Machinery: Fitzgerald,A.E.,CharlesKingsely Jr. Stephen D. Umans, McGraw Hill Books Co., Seventh Edition, 2020.
7. Theory & Performance of Electrical Machines, Gupta J.B, Katsons Books, 2022.
8. NPTEL course as suggested by the department.

Course Code	Course Title	Hours per week L-T-P	Credit C
EE241303	Digital Electronics	3-0-0	3

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Apply Boolean algebra to simplify logic expressions representing digital circuits.
- **CO2:** Design a digital circuit using the minimum number of gates for a given logical expression.
- **CO3:** Analyze and design of basic combinational circuits.
- **CO4:** Analyze and design basic sequential circuits.
- **CO5:** Use logic gates, programmable memory devices and A/D and D/A converters for practical circuit design.

MODULE 1: Digital Fundamentals:(8 Hours)

Representation of binary numbers, octal and hexadecimal numbers, complements, signed binary numbers, floating point numbers, binary codes, code conversion; Boolean expressions and logic gates, truth tables; laws of Boolean algebra, canonical and standard forms- SOP and POS; logic levels and pulse waveforms.

MODULE 2: Simplification of Boolean Functions:(6 Hours)

Karnaugh map, minimal SOP & POS, don't care conditions, multiple output minimization, Quine-McClusky method, determination and selection of prime implicants.

MODULE 3: Combinational Logic circuits:(9 Hours)

Logic synthesis: universal property of NAND and NOR gates; Adders and subtractors, parallel binary adders, magnitude comparator, decoders and encoders, Multiplexer and demultiplexers, parity generators and checkers.

MODULE 4: Sequential Logic Circuits:(9 Hours)

Flip Flops (FFs) and Latches, S-R, J-K, D and T FFs, Excitation table, Triggering of FFs & Latches, one shot and multivibrator. Registers and counters: Definition, Modes of operation of registers and counters, Shift-Registers, Ripple Counters, Synchronous Counters, Ring Counters, Timing Sequences, Design Procedure.

MODULE 5: IC Structures, Digital memory, A/D and D/A converters:(8 Hours)

Introduction to DTL, TTL, ECL, CMOS logic families; ROM, RAM and PLA; Fan-in, Fan-out, power dissipation, propagation delay, Figure of Merit, Noise level; A/D and D/A conversions: D/A Converter- Specifications- weighted resistor type, R-2R Ladder type. A/D Converters – Specifications – Flash type – Successive Approximation type.

Fan-Out, Power dissipation, Propagation delay, figure of merit, noise level.

Textbooks/References:

1. Digital Design: M. Marris Mano, Pearson, Edition of 2018.
2. Logic Design Theory: N N Biswas, Prentice – Hall, Edition of 1992.
3. Digital Fundamental: T L Floyd, Pearson, Edition of 2017
4. Digital Electronics: R. P. Jain, Mc Graw Hill, Edition of 2009.
5. NPTEL course as suggested by the department.

Course Code	Course Title	Hours per week L-T-P	Credit C
EE241304	Electric Circuit Analysis	3-1-0	4

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Apply network theorems for solving electric circuits.
- **CO2:** Apply elementary graph theory for network analysis
- **CO3:** Analyze transient response of various electrical networks.
- **CO4:** Analyze Two-port network characteristics and parameters.
- **CO5:** Synthesize various networks.

MODULE 1: Network theorems for DC, AC and Reduction Circuits:(10 Hours)

Analysis of electrical networks with independent, dependent, ideal, practical voltage and current sources, Millman's theorem and Tellegen's theorem (AC and DC), Reciprocity theorem, Compensation theorem, Maximum Power Transfer Theorem, Thevenin's Theorem and Norton's Theorem for AC circuits; Analysis of coupled circuits; Dot rule and equivalent conductively coupled forms of magnetically coupled circuits.

MODULE 2: Network Topology:(4 Hours)

Elementary Graph theory as applied to electrical networks. Matrices of graph: incidence matrix, tie-set matrix, circuit matrix, cut set matrix; advanced techniques of equation formulation for numerical solutions.

MODULE 3: Transient phenomenon: (8 Hours)

Initial and final conditions in network elements, forcing functions, step and ramp functions. Solution of simple circuits undergoing transient disturbances, A.C. transients, time domain equations and solutions by Laplace transforms.

MODULE 4: Two port Network:(8 Hours)

General principles, relationship of two port variables, impedance parameters, admittance parameters, transmission parameters, hybrid parameters. Relationship among parameters. π network, T network, interconnections of two-port networks.

MODULE 5: Network Synthesis: (10 Hours)

Realizability of networks, positive real function (PRF) and its properties, Hurwitz polynomial, Routh-Hurwitz Array, properties of various immittance functions (LC, RC, RL), Foster and Cauer forms of realization of network for a given driving point impedance/admittance function.

Textbooks/References:

1. Circuit Theory Analysis and Synthesis: Abhijit Chakravarti, Dhanpat Rai & Comp, 2021.
2. Network Analysis and Synthesis: A. Anand Kumar, PHL Learning Pvt. Ltd, 2019.
3. Electric Circuit: J.A. Edminster, McGraw Hill, 2017.
4. Networks and Systems: D Roy Chowdhury, New Age, 2013.
5. A course in Electric Circuit Analysis: M.L Soni, Dhanpat Rai Publication, 2016.
6. Network Analysis, M E Van Valkenburg, Pearson Education, 2019.
7. Engineering Circuit analysis, W. H. Hayt Jr. & J.E. Kemmerly, McGraw Hill, 2013.
8. NPTEL course as suggested by the department.

Course Code	Course Title	Hours per week L-T-P	Credit C
MA241305C	Mathematics III-C	3-0-0	3

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Apply Laplace transform techniques for solution of engineering problems
- **CO2:** Solve the problems of vector spaces and Linear transformations
- **CO3:** Solve engineering problems using numerical methods
- **CO4:** Apply probability and statistical techniques for solution of engineering problems

MODULE 1: Laplace Transforms: (7 Hours)

Definition and properties of Laplace Transform and Inverse Laplace Transform, Laplace Transform and Inverse Laplace Transform of derivatives and integrals, Method of finding inverse Laplace Transform using partial fractions, Convolution Theorem, Solution of ordinary differential equations using Laplace Transforms.

MODULE 2: Linear Algebra: (7 Hours)

Linear Systems of Equations: Existence and Uniqueness of solutions and Dimension of the Solution Space associated with the system of linear equations; Vector Spaces: Definition of Vector Space, Sub-spaces, Linearly Independence and Dependence, Linear Span, Basis, Dimension

Row, Line and Column space of a matrix.

Basis and Dimension: Definition and examples, Linear dependence and independence, Dimension of a subspace. Linear transformations: Definition and examples, the Kernel of a Linear Transformation, Matrix representation of linear transformation and systems of linear equations, Rank-nullity theorem.

MODULE 3: Numerical Methods: (14 Hours)

Errors in Numerical Calculations, Solution of Algebraic and Transcendental Equations: Bisection Method, Iteration Method, Newton-Raphson Method. Interpolation: Newton Forward, Backward and Lagrange interpolation, Numerical Differentiation and Integration: Trapezoidal and Simpson's rule. Solution of Linear Systems: Gauss elimination and pivoting, Gauss-Seidel methods, Numerical Solution of Ordinary Differential Equations: Euler's and Runge-Kutta Methods.

MODULE 4: Probability and Statistics: (12 Hours)

Basic Probability Theorems, Conditional Probability and Baye's theorem; Random Variables, Probability Distributions; Mean and Variance, Median and Mode; Binomial, Poisson, Gaussian and Normal distribution and their applications; Joint distribution: Covariance and Correlation; Random Sampling, Estimation of Parameters: Point Estimate, Interval Estimate, Maximum Likelihood Method; Confidence Intervals, Testing of Hypothesis, Goodness of Fit Test, Curve Fitting: Least Square Method.

Textbooks/References:

1. Advanced Engineering Mathematics: Erwin Kreyszig, John Wiley, 2011.
2. Introduction to Linear Algebra: Strang G., Wellesley MA, Cambridge Press, 2016.
3. Numerical Methods for Scientific and Engineering Computation: Jain M.K., Iyengar, S. R. K., Jain, R. K., New Age International (P) Ltd., 8th Edition, 2022.
4. Introduction to Probability and Statistics for Engineers and Scientists: Ross, S., Pearson, 6th Edition, 2021.

Course Code	Course Title	Hours per week L-T-P	Credit C
HS241306	Indian Knowledge Systems	2-0-0	2

Course Objectives:

1. To introduce students to the foundational concepts of Indian Knowledge Systems (IKS).
2. To explore the contributions of ancient and medieval India in science, technology, mathematics, and philosophy.
3. To develop an appreciation for traditional Indian knowledge and its relevance in modern contexts.
4. To encourage critical thinking about indigenous knowledge systems and their applications.

Course Outcome (CO):

After completion of the course, the students will be able to

- **CO1:** Understand the key principles of Indian Knowledge Systems.
- **CO2:** Recognize India's contributions to science, mathematics, medicine, and engineering.
- **CO3:** Analyze the interdisciplinary nature of IKS and its connections with modern technology.
- **CO4:** Appreciate the philosophical and ethical dimensions of traditional Indian knowledge.

MODULE 1: Introduction to Indian Knowledge System (IKS) (4 Lectures)

- Definition, scope, and significance of IKS
- Sources of Indian knowledge: Vedas, Upanishads, Puranas, Smritis, and other texts
- Classification of knowledge in Indian tradition (Vidyas and Kalas)

MODULE 2: Science & Technology in Ancient India (12 Lectures)

- Contributions to mathematics (Zero, Decimal System, Vedic Mathematics, Aryabhata, Brahmagupta)
- Astronomy & Space Science (Surya Siddhanta, Planetary Motion, Concept of Time)
- Metallurgy and material sciences (Iron Pillar, Wootz Steel, Ayurvedic alloys)
- Civil engineering & architecture (Temple architecture, Town planning, Water management)

MODULE 3: Indian Philosophical & Ethical Systems (6 Lectures)

- Key schools of Indian philosophy (Nyaya, Vaisheshika, Samkhya, Yoga, Vedanta)
- Concept of Dharma, Karma, and sustainable living
- Ethics in science and technology from an Indian perspective

MODULE 4: Traditional Medicine & Wellness Systems (4 Lectures)

- Foundations of Ayurveda, Yoga, and Siddha
- Holistic health and wellness practices
- Relevance of traditional medicine in modern healthcare

MODULE 5: IKS in Contemporary Applications (4 Lectures)

- Revival of traditional knowledge in modern science and engineering
- Case studies (e.g., Panchakarma in wellness tech, Vastu in architecture)
- Government initiatives (National Education Policy 2020, IKS Division under AICTE)

Text Books:

1. "Indian Knowledge Systems" – Kapil Kapoor & Avadhesh Kumar Singh
2. "Ancient Indian Leaps into Mathematics" – B.S. Yadav
3. "Science and Technology in Ancient India" – O.P. Jaggi
4. "The Positive Sciences of the Ancient Hindus" – Brajendranath Seal
5. AICTE Model Curriculum for IKS (Available online)

Online Resources:

- NPTEL / SWAYAM courses on IKS
- AICTE-IKS Portal (<https://iksindia.org/>)
- Digital Library of India (<https://dli.gov.in>)

Course Code	Course Title	Hours per week L-T-P	Credit C
AC241307	Language other than MIL	1-1-0	0

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Demonstrate a clear understanding of basic grammar rules including subject-verb agreement, tenses, and cases, and apply them in constructing grammatically correct sentences.
- **CO2:** Identify and correct common grammatical and usage errors in written English, enhancing sentence effectiveness and clarity.
- **CO3:** Enhance their vocabulary through the use of synonyms, antonyms, idiomatic expressions, and phrasal verbs, improving both written and spoken communication.
- **CO4:** Apply verbal reasoning skills to comprehend, arrange, and complete sentences and paragraphs effectively.
- **CO5:** Develop strong reading comprehension skills using skimming, scanning, and intensive reading techniques to extract key information from various texts.
- **CO6:** Produce coherent and concise writing through precis writing and essay composition, demonstrating logical flow, clarity, and appropriate structure.

MODULE 1: Basic grammar usage

(10 Lectures)

Subject-verb agreement, narration, tenses

Case: Subjective case, possessive case, objective case

Writing effective sentences, Spotting errors

MODULE 2: Vocabulary Building

(6 Lectures)

Synonyms and Antonyms, One word substitution,

Idioms and Phrases and Phrasal Verbs

MODULE 3: Verbal ability

(8 Lectures)

Cloze test

Sentence rearrangement, Sentence completion,

Paragraph Jumbles

MODULE 4: Writing and Reading Ability

(8 Lectures)

Reading subskills, Skimming, Scanning and Intensive reading

Reading Comprehension, Precis Writing, Essay Writing

Text Books

1. Objective General English: S.P.Bakshi, Arihant Publication India Ltd., 2021
2. Descriptive English: S.P.Bakshi, Arihant Publication India Ltd., 2016

Course Code	Course Title	Hours per week L-T-P	Credit C
EEE241311	Basic Electronics Lab	0-0-2	1

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Test the working of different diodes, transistors, CRO probes and measuring instruments.
- **CO2:** Design the circuits with basic semiconductor devices
- **CO3:** Measure and record the experimental data, analyze the results, and prepare a formal laboratory report.

Part I (Virtual Lab)

1. To study V-I Characteristics of Zener diode and Zener diode as a voltage regulator. **(2 Hours)**
2. Input and output characteristics of transistor CB configuration. **(2 Hours)**
3. Input and output characteristics of transistor CE configuration. **(2 Hours)**
4. Studies of JFET, MOSFET characteristics. **(2 Hours)**
5. Studies of basic properties of Operational Amplifier: Inverting and Non-Inverting Amplifiers. **(2 Hours)**

Part II (Hardware Experiments)

1. Implement rectifiers using p-n junction diodes and study the effect of filters. **(2 Hours)**
2. Input and output characteristics of a BJT CB configuration. **(2 Hours)**
3. Input and output characteristics of a BJT CE configuration. **(2 Hours)**
4. To study Half-Wave & Full-Wave rectifier with and without filter **(2 Hours)**
5. Comparison of Performance of Self Bias and Fixed Bias Circuits **(2 Hours)**

Course Code	Course Title	Hours per week L-T-P	Credit C
EEE241312	Electrical Machines-I Lab	0-0-2	1

Prerequisites: Basic Electrical Engineering

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Familiarisation with the mode of starting, switching-off, and taking precautionary measures while handling electrical machines
- **CO2:** Apply the theoretical concepts for experimentation with different electrical machines
- **CO3:** Develop report writing skills

LIST OF EXPERIMENTS

1. O.C.C. of D.C. Generators. **(3 Hours)**
2. Load Test on D.C. Series Generators. **(3 Hours)**
3. Load Test on D.C. Shunt Generators. **(3 Hours)**
4. Starting up and speed control of D.C. Shunt Motors. **(3 Hours)**
5. Open circuit and short circuit test on single-phase Transformers. **(3 Hours)**
6. Sumpner's Test or back-to-back test on two similar single-phase Transformers. **(3 Hours)**

Course Code	Course Title	Hours per week L-T-P	Credit C
EE241313	Digital Electronics Lab	0-0-2	1

Course Outcome (CO): After completion of the course, the students will be able to

- **CO1:** Realize basic gates and verify their truth tables
- **CO2:** Perform experimental analysis of combinational and sequential logic circuits.
- **CO3:** Write an engineering report in an organized way

LIST OF EXPERIMENTS

1. Realization of basic gates by using universal gates and verification of their truth tables. **(2 Hours)**
2. Realization of half adder and full adder circuits by using basic gates. **(2 Hours)**
3. Combinational Logic Design using 74xx ICs. **(2 Hours)**
4. Arithmetic Circuit construction and testing using 74xx ICs: Adders. **(2 Hours)**
5. Construction of 1-bit comparator using 74xx ICs. **(2 Hours)**
6. Code converters – Binary to Gray and Gray to binary using basic gates. **(2 Hours)**
7. Verification of Truth Table of SR Flip-Flop using NOR and NAND gates. **(2 Hours)**
8. Verification of Truth Tables of JK, D, T Flip-Flops using 74xx ICs. **(2 Hours)**
9. Construction of 3-Bit synchronous counter using flip flops. **(2 Hours)**
10. Construction of Decade Counter using 74xx ICs. **(2 Hours)**

Suggested Books and Other Study Materials

1. Digital Design: M. Marris Mano, Pearson, 1984.
2. Digital Electronics Practice using Integrated Circuit: R. P. Jain, McGraw Hill Education, 2009.
